

GlobalHack VI Software Competition

October 21-23, 2016

LTU Quest Project

Team Members:

Nicholas Paul; *Senior; CS Scientific Software, Math Minor, Team Captain*

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Faculty Advisor: Dr. CJ Chung

The LTU MCS GlobalHack team had a great time this weekend at GlobalHack. We learned so much about working in teams, developing useful software, and how we can make a difference in the homeless community.

The Competition

GlobalHack, <https://globalhack.org>, is a software competition where teams compete to develop software from the ground up over the course of one weekend. This year's competition was held in St. Louis and consisted of over 1000 participants (150+ teams) all competing to develop software to end or prevent homelessness.

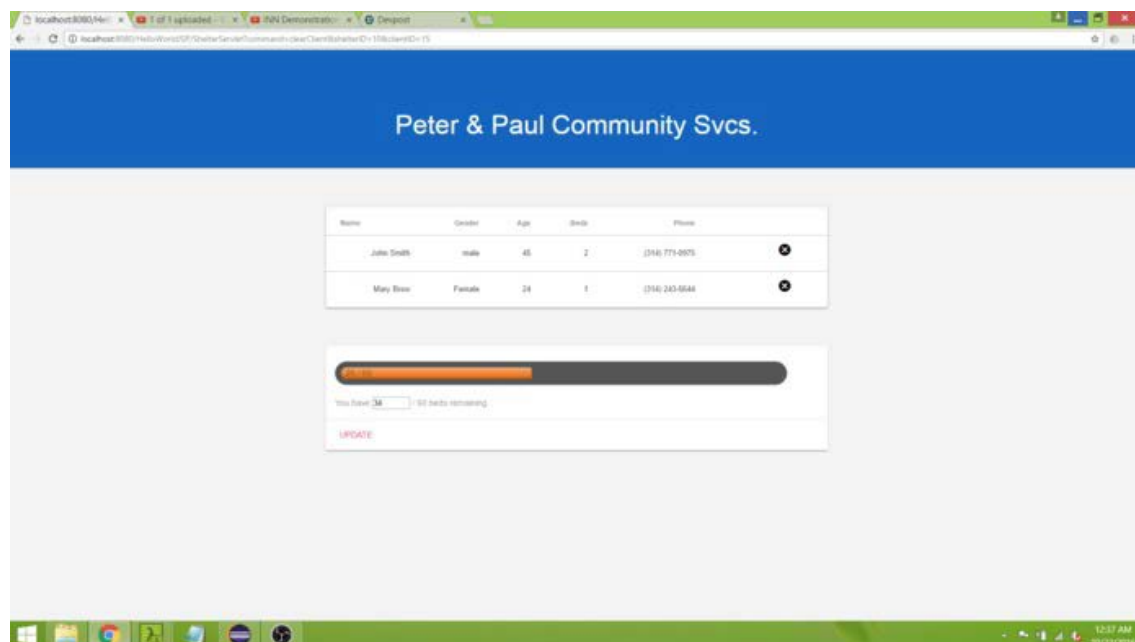


GlobalHack VI consisted of 1000+ participants making up 150+ teams

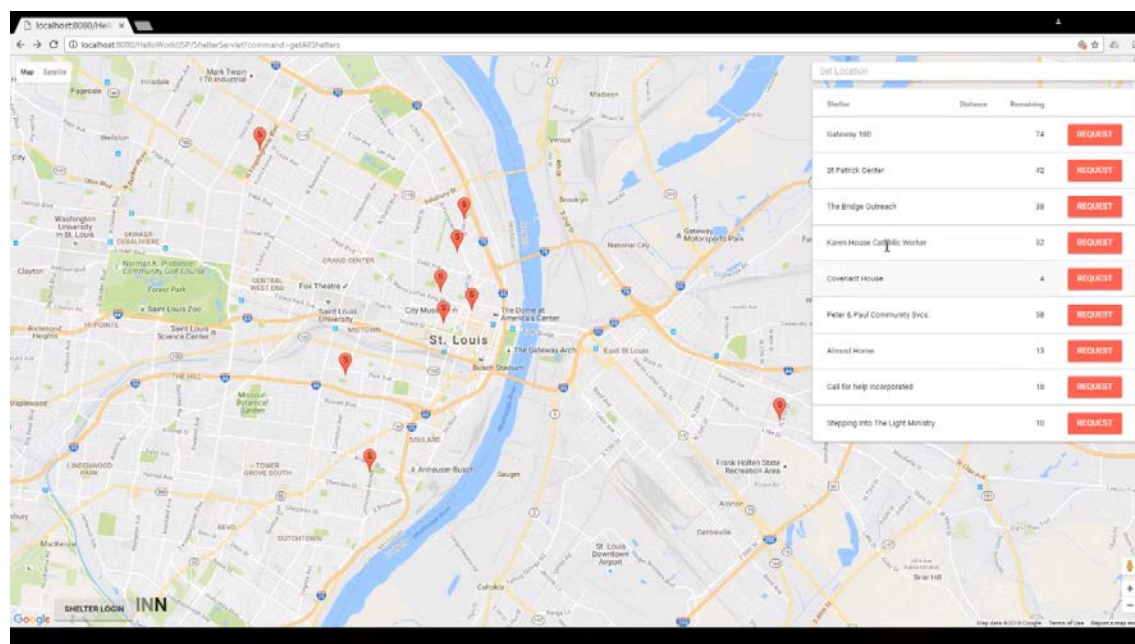
Our Project

The project was centered around the problem of intake management. Currently, there is no system in place for organizing the distribution of homeless individuals across the many shelters in large cities. Additionally, there is no way for homeless individuals to know which shelter will best meet their needs without extensive planning or last-minute visitations. During a homelessness panel, our team discovered that most homeless individuals have access to the internet via smartphone or computer. Therefore, we also wanted to make our solution available to homeless individuals themselves.

First, the system implements Google Maps API to allow shelters to quickly redirect homeless individuals to nearby shelters. Second, it provides a way for homeless individuals to communicate with the shelters via online forms. Finally, it provides a dashboard interface for shelters so that they can manage the number of individuals staying in their beds.



The shelter dashboard displays the current occupancy level and all pending requests



The map interface allows homeless individuals and shelter offices to find available beds

The project's back-end system was written in Java and JavaServer Pages and hosted on an Apache Tomcat Server. We used Oracle's MySQL database system to store data regarding clients and shelters. The front-end interface was designed using HTML5, CSS, and JavaScript. We took advantage of Google's Material Design libraries to give the website its elegant and clean look. The mapping interface uses Google Maps API and jQuery to load and display the shelters and compute distances.

Results

The project made it to the finals along with about 20% of competing teams. Unfortunately, it did not place in the top 5 and only the top 5 teams from each division were awarded a prize.

We received the prompt Friday night and submitted it very late Saturday evening. In just 17 hours we were able to successfully implement all of the core features we wanted the system to have. The project totaled nearly 2000 lines of Java, JSP, HTML, CSS, and JavaScript. This means that the team averaged about 1 line of code per minute per member – an exceptionally high rate. We also spent several hours planning, attending panels and designing the system before we began coding.

Overall, we are very satisfied with the project we worked on and we are happy that we were able to contribute to the problem of homelessness. The team plans on publishing the source code with a free, open source license so that anyone, including homeless shelters, may use the software. The project submission is publicly available at <https://devpost.com/software/globalhack>

What We Learned

This experience taught us things that we could have never learned in a classroom setting. Working in a fast paced, highly competitive, team based environment allowed us to experience software development in the "real-world." We were forced to stay organized, work quickly and exercise extensive creativity lest we fall behind and lose the opportunity to place in the competition.



From Left to Right: Nicholas Paul, Jacob Crane, and Nick Virag

Overall, GlobalHack was a great experience and the entire team is very appreciative of Lawrence Technological University for giving us the opportunity to participate in such a great event. We are very proud of our work and we hope that it will be used to help aid in the problem of homelessness.

Acknowledgements

The team is very grateful for the opportunity to attend such a huge event. This would not have been possible without the support from the Quest program. We appreciate Dean Hsiao-Ping Moore for launching such a great program. We also want to thank Quest director Dr. Shannon Timmons for helping us get funding from the program, Marilyn Wiseman for all her work on the car rental process and Tracy Kash for process reimbursement. Finally, none of this would have been possible without the support and encouragement of Dr. Patrick Nelson, Dept. Chair and our faculty advisor Dr. CJ Chung.